

5.2 Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence

Lesson Introduction

 Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.5.1 Construct a copy of a segment or an angle.	 Learning Targets	- I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent. - I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent. - I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.	
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I use constructions to determine if SAS, ASA, and SSA congruence can be used to show that two triangles are congruent? - How can I use given information to determine if SSS, SAS, or ASA can be used to show that two triangles are congruent?			
 Lesson Narrative	In this lesson, students will explore congruent relationships using side lengths and specified angles to determine if two triangles are congruent. Students will use their explorations with constructions to discover the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem. They will use this knowledge to determine if two triangles are congruent.			
 Component Summary	5.2.1 Warm-Up (~5 min.)	Geometry in art (<i>SE p. 230</i>)	5.2.3 Bring It Together (10-15 min.)	Formalized introduction to the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem (<i>SE p. 232</i>)
	5.2.2 Exploration (20-25 min.)	Exploring various constructions to discover congruent relationships between triangles (<i>SE p. 230</i>)	5.2.4 Wrap-Up (~5 min.)	Practice using the triangle congruence criterion to determine if triangles are congruent (<i>SE p. 233</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Side-Angle-Side (SAS) Congruence Postulate - Angle-Side-Angle (ASA) Congruence Theorem <u>Additional Terms:</u> - Angles - Congruence		- Compass - Ruler - Patty Paper	The design of the lesson is not to give students the information, but for them to discover the SAS Congruence Postulate and ASA Congruence Theorem. Consider doing the exploration on your own to see how this unfolds to make it easier to facilitate with students.

 Teacher Tips	Common Misconceptions - Students may think that SSA can be used to determine congruency between two triangles. - Students may forget the “included” component of SAS and ASA. - Students may misread side lengths using line segment notation.	Things to Consider The majority of this lesson is 5.2.2 Exploration. Students will be using tools, such as a compass and patty paper explicitly during the activity.
	Literacy and Language Routines Think-Pair-Share For 5.2.2 Exploration, students will be working together to discover congruent relationships between sides and angles. Students will get the opportunity to work on their own, pair up with a partner, and share their work with one another to come to a consensus.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection In 5.2.3 Bring It Together, students will be making conjectures based on what they learned through their initial exploration. Students will create arguments based on what they explored in the prior activity.
	 Differentiate Instruction - Consider pairing students based on their performance in the prior lesson. The data from 5.1.4 Wrap-Up from Lesson 1 can serve as data for student groupings for this lesson. - In 5.2.3 Bring It Together question 2, have students mark the congruent parts on the given triangles. Then, have the student write the words “side” or “angle” next to the marked part. This move will help students visualize which congruence postulate/theorem is represented in the problem. 5.2.4 Wrap-Up provides students with an opportunity to demonstrate mastery of determining if two triangles are congruent, including what theorem or postulate was used to make that determination. Consider having students list what side/angle relationships are valid for determining congruence of triangles and what relationships aren’t for an additional check for understanding.	
Lesson Notes:		

5.2.1 Warm-Up: Geometry in Art

Component Overview: Students will explore geometry in art through the largest Pysanka. This activity is designed to be done independently.



5.2.1 Warm-Up: Geometry in Art

A Pysanka is a decorated egg that is traditional in Ukraine. The decoration on the eggs is created using a wax resist (batik) method. Families often have their own unique pattern. The largest Pysanka is in Vegreville, Canada. It weighs 5,000 pounds, is 31 feet (ft) tall, and 18 ft wide. A traditional Pysanka egg is 2.5 inches tall and 1.75 inches wide.



1. What do you notice about Pysanka eggs?

Answers will vary. The Pysanka egg is large. I see that the egg has been decorated with many shapes.

2. Estimate the number of traditional Pysanka eggs that will fit into the largest Pysanka. Show your work or explain how you found your estimate.

Answers will vary. Estimated values: Traditional egg 3 inches tall and 2 inches wide. This large egg is 375 inches tall and 220 inches wide. Using the tall dimension $375 / 3 = 125$, using the wide dimension $220 / 2 = 110$. To estimate, I should use the smaller number, so I estimate that about 110 traditional Pysanka eggs will fit inside the largest Pysanka egg.



Allow students to share their answers to question 2 with the entire class. Encourage students to agree or disagree with student estimates as viable or not viable.

5.2.2 Exploration: Congruence Relationships

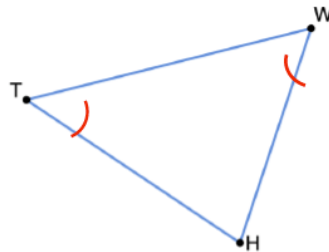
Component Overview: Students will complete various constructions to discover congruent relationships between triangles. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss, in order to ensure students are on the right track prior to progressing through the next part of the activity.



5.2.2 Exploration: Congruence Relationships

1. Triangle TWH is shown.

- a. Construct $\triangle XMP$ so that $\overline{XM} \cong \overline{TW}$, $\overline{PM} \cong \overline{HW}$, and $\angle TWH \cong \angle PMX$ using the given steps.



Construction	Steps
	1. Copy $\overline{TW}$ using a compass to construct $\overline{XM}$ using the orange line so that $\overline{XM} \cong \overline{TW}$.
	2. Copy $\angle TWH$ to construct $\angle PMX$ so that $\angle TWH \cong \angle PMX$.
	3. Copy $\overline{HW}$ to construct $\overline{PM}$ so that $\overline{PM} \cong \overline{HW}$.
	4. Draw $\overline{PX}$.

- b. Use patty paper to determine if $\triangle XMP$ is congruent to $\triangle TWH$.
- c. Discuss the following with your partner: If two sides of a triangle and their included angles are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

Answers will vary. If two sides of a triangle and their included angle are congruent, then all parts of the triangles are congruent. After using the patty paper, I can see that the remaining side is congruent to its corresponding side and the other two angles are congruent to their corresponding angles. This makes all sides and angles of the triangle congruent to the corresponding parts.



For this activity, students will need the following materials:

- Compass
- Ruler
- Protractor
- Patty Paper



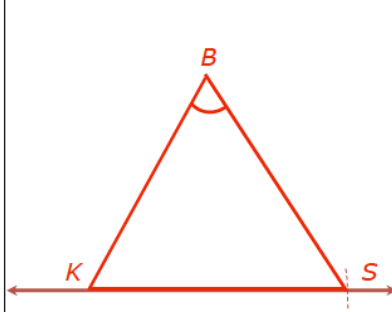
For each question, consider using technology, such as a document camera or GeoGebra to illustrate the steps of the construction so that students can follow along in review.



- When constructing $\angle PMX$, how long do you draw $\overline{MP}$?
- How can you use patty paper to determine if all parts of $\triangle XMP$ are congruent to $\triangle TWH$?
- What does “included angle” mean?

5.2.2 Exploration: Congruence Relationships (continued)

2. Construct $\triangle KSB$ so that $\overline{KS} \cong \overline{TH}$, $\overline{SB} \cong \overline{TW}$, and $\angle TWH \cong \angle SBK$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{KS}$ using the red line so that $\overline{TH} \cong \overline{KS}$.
	2. Copy $\overline{TW}$ to construct $\overline{SB}$ so that $\overline{TW} \cong \overline{SB}$.
	3. Copy $\angle TWH$ to construct $\angle SBK$ so that $\angle TWH \cong \angle SBK$.
	4. Draw $\overline{BK}$.

- a. Use patty paper to determine if $\triangle KSB$ is congruent to $\triangle HTW$.
- b. Discuss the following with your partner: If two sides of a triangle and a non-included angle are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

Answers will vary. No, it does not. If two sides of a triangle and a non-included angle are congruent, then it cannot be determined if all parts of the triangle are congruent. The non-included angle doesn't provide any information about the 3rd side length and the triangle can be created different ways using two sides and a non-included angle.



The $\triangle TWH$ is not on every question in the student workbook. Consider having students use patty paper to cut down on having to flip to previous pages when working through the additional questions. Students may struggle with knowing what angle to draw $\overline{SB}$ at point S . Resist giving them help here, as this is a key part of determining that SSA is not a valid statement of congruence.



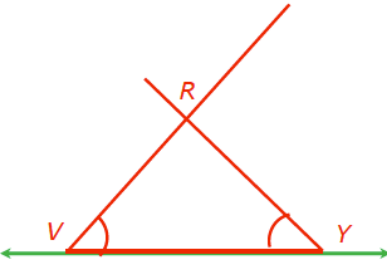
Why do you think that if two sides of a triangle and a non-included angle are congruent then we can't know for certain that all parts of the triangles are congruent too?



Why do you think the included vs. non-included angles would make a difference in the determination of whether not two triangles are congruent?

5.2.2 Exploration: Congruence Relationships (continued)

3. Construct $\triangle RYV$ so that $\overline{VY} \cong \overline{TH}$, $\angle WTH \cong \angle RYV$, and $\angle WHT \cong \angle RYV$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{VY}$ using the green line so that $\overline{TH} \cong \overline{VY}$.
	2. Copy $\angle WHT$ to construct $\angle RYV$ so that $\angle WHT \cong \angle RYV$.
	3. Copy $\angle WTH$ to construct $\angle RYV$ so that $\angle WTH \cong \angle RYV$.
	4. Label the intersection of the sides of the two angles point R.

- Use patty paper to determine if $\triangle RYV$ is congruent to $\triangle WHT$.
- Discuss the following with your partner: If two angles of a triangle and their included side are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

Answers will vary. If the congruent side is between the congruent angles, all parts of the triangle will be congruent. With this triangle construction, there will be no other way to create a triangle so it must be congruent.



When reviewing the answers, consider having a student come up to the board to demonstrate the steps of the construction in question 3.



How long should the rays of the copied angles be drawn?



- What are the similarities and differences between the three triangles you constructed in this activity?
- Is there any relationship between the information provided for each triangle?
- What do you notice about these three?
- What do you wonder about them?

5.2.3 Bring It Together: Triangle Congruence

Component Overview: Students will connect what they discovered in 5.2.2 Exploration to a formalized introduction to the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem.



5.2.3 Bring It Together: Triangle Congruence

1. Complete the statements.

- a. $\triangle XMP$ was created using

☐ 1
☒ 2
☐ 3

 side(s) and

☒ 1
☐ 2
☐ 3

 angle(s).

The construction showed that it is

☒ possible
☐ not possible

to assume that if the

☐ 1
☒ 2
☐ 3

 side(s) of a triangle

and their

☐ adjacent angle(s)
☒ included angle(s)
☐ opposite angle(s)

 are congruent,

then the triangles are congruent.

- b. $\triangle KSB$ was created using

☐ 1
☒ 2
☐ 3

 side(s) and

☒ 1
☐ 2
☐ 3

 angle(s). The construction showed that it is

☐ possible
☒ not possible

 to assume that if the

☐ 1
☒ 2
☐ 3

side(s) of a triangle and the

☐ included angle(s)
☒ non-included angle(s)

are congruent, then the triangles are congruent.



Consider modeling question 1a as a think-aloud with the class. Students often need support on how to think through fill-in-the-blank style questions.



For question 1b, consider having students review their answers as a class prior to moving forward to question 1c. This way, students can engage in question 1c as an independent practice for themselves.



What are the differences between what you discovered in question 1a and 1b?

5.2.3 Bring It Together: Triangle Congruence (continued)

c. $\triangle RYV$ was created using ☐ 1 ☐ 2 ☐ 3 side(s) and ☐ 1 ☐ 2 ☐ 3 angle(s).

The construction showed that it is ☐ possible ☐ not possible

to assume that if the ☐ 1 ☐ 2 ☐ 3 side(s) of a triangle

and its ☐ adjacent angle(s) ☐ included angle(s) ☐ opposite angle(s) are congruent,

then the triangles are congruent.



Side-Angle-Side (SAS) Congruence Postulate

If two sides and the included angle of one triangle are congruent to two sides and the included angle of a second triangle, respectively, then the two triangles are congruent.



Angle-Side-Angle (ASA) Congruence Theorem

If two angles and an included side of one triangle are congruent respectively to two angles and an included side of a second triangle, then the two triangles are congruent.



What makes the triangle in question 1c similar to the ones in questions 1a and 1b? What are some of the differences?



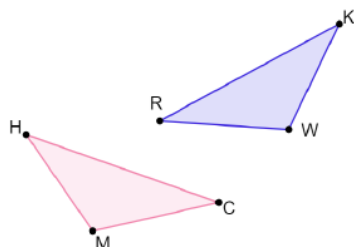
When reviewing the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem, consider reviewing them within the context of the work students have already done. For example, don't simply give the postulate, but connect it to the corresponding triangle from the prior three they have worked with as well.



Why do you think that there is an SAS Congruence Postulate and an ASA Congruence Theorem but no congruence postulate or theorem for SSA?

5.2.3 Bring It Together: Triangle Congruence (continued)

2. Triangles HCM and RKW are shown. Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate to show that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{MC} \cong \overline{RW}$, $\overline{HC} \cong \overline{KR}$, $\angle MCH \cong \angle WRK$	Yes	SAS
$\overline{MC} \cong \overline{KW}$, $\overline{HM} \cong \overline{WR}$, $\overline{HC} \cong \overline{KR}$	Yes	SSS
$\overline{KW} \cong \overline{HM}$, $\angle KRW \cong \angle CHM$, $\overline{MC} \cong \overline{RW}$	No	
$\overline{RK} \cong \overline{CH}$, $\angle MCH \cong \angle WRK$, $\angle MHC \cong \angle RKW$	Yes	ASA
$\angle RKW \cong \angle CHM$, $\angle RWK \cong \angle CMH$, $\overline{KW} \cong \overline{HM}$	Yes	ASA



For each line of the chart, students can mark the congruent parts on the given triangles. Then, have the student write the words “side” or “angle” next to the marked part. This move will help students visualize which congruence postulate/theorem is represented in the problem. Teachers can also use this step to check for where misunderstandings occur when developing remediation plans.



To add some physical movement to this activity, consider labeling each corner of the room with the following:

- SAS
- ASA
- SSS
- SSA

Display each row of the table and have students choose a corner to determine which one they agree with. Have students share out their rationale for their choice.



After completing the chart, ask students why SSA was never an answer for the third column.



For a challenge, have students create two triangles that could be proved to be congruent by a specific theorem or postulate.

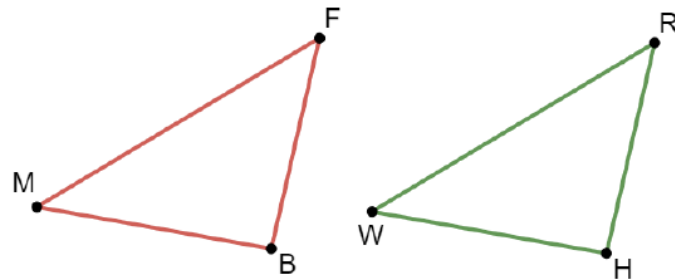
5.2.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will use the triangle congruence criterion they have learned to determine if triangles are congruent. This activity should be done independently to determine mastery of the content of this lesson.



5.2.4 Wrap-Up: Ticket Out the Door

Triangles BFM and WRH are shown.



This question assesses student understanding of the Side-Side-Side (SSS) Congruence Postulate and Side-Angle-Side (SAS) Congruence Postulate. The data from the student responses can provide insight on whether the students need additional support on the content from this lesson.

Consider having students list what side/angle relationships are valid for determining congruence of triangles and what relationships are not, for an additional check for understanding.

- Determine if the given information is sufficient to determine that the triangles are congruent. If so, state the congruence theorem or postulate to show that.

Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{FB} \cong \overline{HW}$, $\overline{FM} \cong \overline{RW}$, $\angle BFM \cong \angle HWR$	Yes	SAS
$\overline{FB} \cong \overline{RH}$, $\angle FBM \cong \angle RHW$, $\overline{MB} \cong \overline{WH}$	Yes	SAS
$\overline{FB} \cong \overline{HW}$, $\overline{BM} \cong \overline{HR}$, $\overline{FM} \cong \overline{RW}$	Yes	SSS